

Introduction to JSON

[Video](#) about this material.

JSON, pronounced as in the name “Jason”, stands for “JavaScript Object Notation”. It is used to store data in configuration files in applications and storing data in games for example. JSON is used in web applications for sending data between clients (like web browsers) and servers. The other common data format is XML.

In this week’s distributed data programming assignment, we will see another use of the JSON in the APIs for several web services. An API is a set of rules and specifications that allows different software systems to communicate and interact with each other. In this project we will see how we can use an API to get data from a database stored on another computer system and find the result we are looking for. Libraries of code for dealing with JSON formatted data exists in many languages. We will be using one of these libraries for the project this week.

Here’s some basic information about JSON that will give you enough knowledge to be able to work with JSON formatted data. To obtain the sample files, use the following command in your terminal window:

```
git clone https://github.com/Casper-College-COSC/JSON-python
```

Here’s an example of what JSON looks like:

```
{
  "name": "Alice",
  "age": 30,
  "isStudent": false
}
```

Notice the following details about this notation:

1. The information is in **key/values** pairs. The **key** is on the left side of the colon, “:”, and the **value** is on the other side, so in the above example, “name” is a key and “Alice” is a value.
2. The key/value pairs are separated by commas.
3. The object is enclosed by curly braces “{ }”
4. A key must be a string, enclosed by double quotes.
5. The value can be a string, number, a Boolean, an array, another object, or null

Here’s an example that has an array in it:

```
{
  "name": "Bob",
```

```
"hobbies": ["reading", "cycling", "chess"]
}
```

In the above example, the key “hobbies” has an associated value that is an **array** containing strings. Note that each element in the array is separated by a comma.

Accessing the data stored in this format

The main idea with data in JSON format is that you use the key (which is fixed and known) to get to the data which is variable. Many languages have libraries that can easily read in JSON formatted text data into some data structure. Take a look at the following example in Python (note that it doesn’t matter if you haven’t had any experience with Python to be able to understand what this code is doing.) You are encouraged to try this example on your own so you can elaborate on it for this assignment. The `import json` line should work with any python environment. This library is part of Python’s standard library.

Given a text file called `json1.txt` that contains the text from the above example:

```
{
  "name": "Bob",
  "hobbies": ["reading", "gaming"]
}
```

The following Python programs (`json_example.py`) opens the file, reads the contents into a Python dictionary, then accesses the value at the “hobbies” key using the indexing notation with the square brackets “[]”. To run this code yourself, use a codehs

```
import json

with open("json1.txt", "r") as file:
    data = json.load(file)

print(data)
print(data["hobbies"]) # Output: ['reading', 'gaming']
```

Note that the “hobbies” key returns an array. If we wanted to access the 0th element of that array and print out the contents, we could change the last line to:

```
print(data["hobbies"][0])
```

Notice that to access array elements the square brackets are used.

Most json records contain more than one object and as mentioned above, the values in the key/value pairs can also be objects. Take a look at the following example (json2.txt):

```
{
  "person": {
    "firstName": "Eve",
    "lastName": "Smith",
    "address": {
      "city": "New York",
      "zip": "10001"
    }
  }
}
```

1. How would you access the value that has "firstName" as the key?
2. How would you access the value that has the key "city"?

Take a look at this json record (json3.txt):

```
{
  "places" : ["Arizona", "New York", "Georgia", "Texas"],

  "people": [
    {
      "name": "Bob",
      "hobbies": ["reading", "gaming"]
    },
    {
      "name": "Mary",
      "hobbies": ["hiking", "sewing"]
    },
    {
      "name": "Juan",
      "hobbies": ["archery", "skiing"]
    }
  ]
}
```

3. Write the python code to access print the name and the first hobby for each of the items in the "people" array.

TURN IN: The answers to the three questions.

References

https://www.w3schools.com/js/js_json_intro.asp

<https://www.geeksforgeeks.org/javascript/json/>